Slovenská Technická Univerzita – Fakulta elektrotechniky a informatiky, Ilkovičova 3, 812 19 Bratislava

Optimalizácia rojom £astíc Particle Swarm Optimization

Umelá inteligencia 2, zadanie č. 4

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Ročník štúdia: **1. Ing.**
Študijný program: **Robotika a kybernetika**

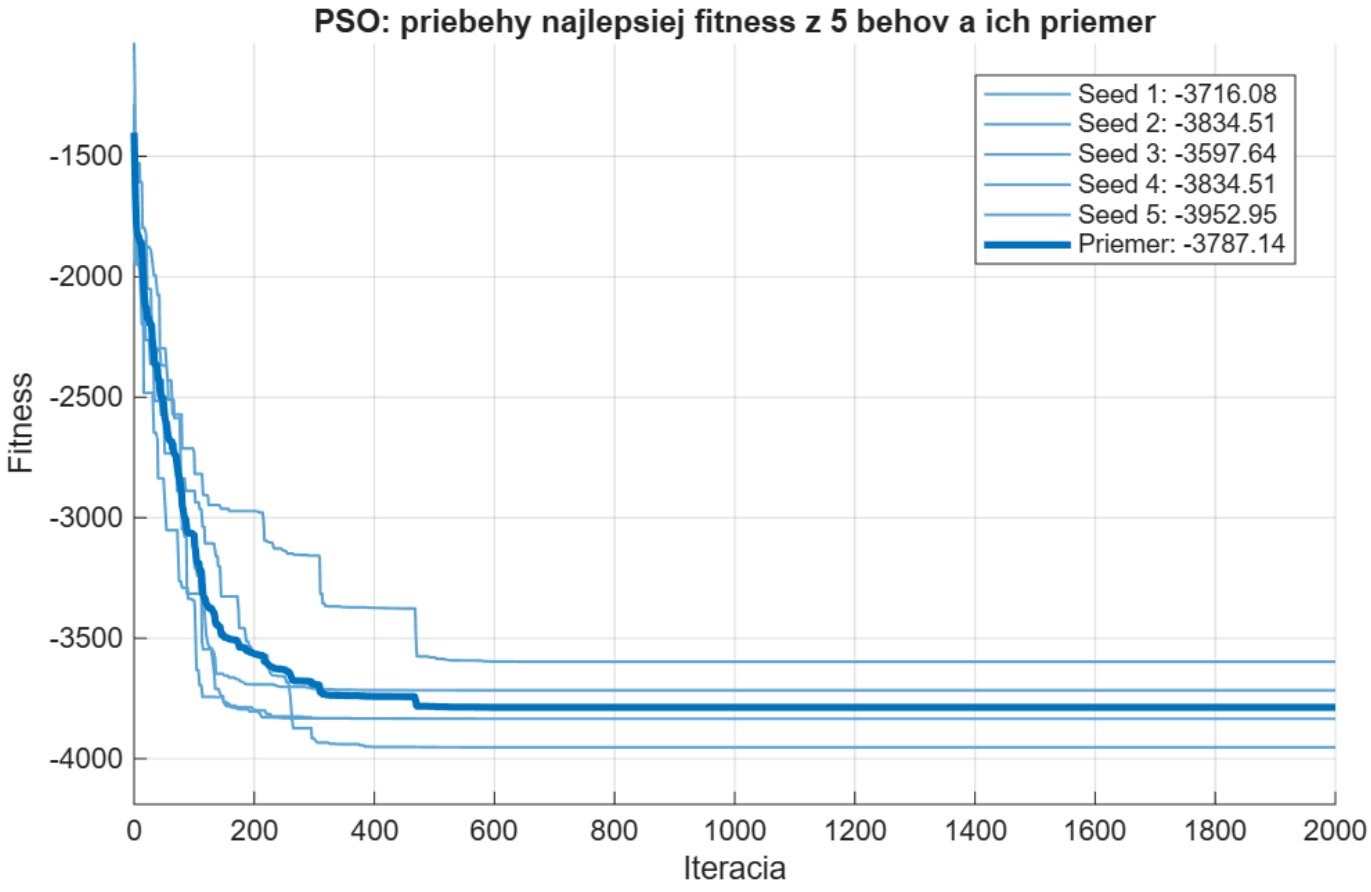
Nastavenia:

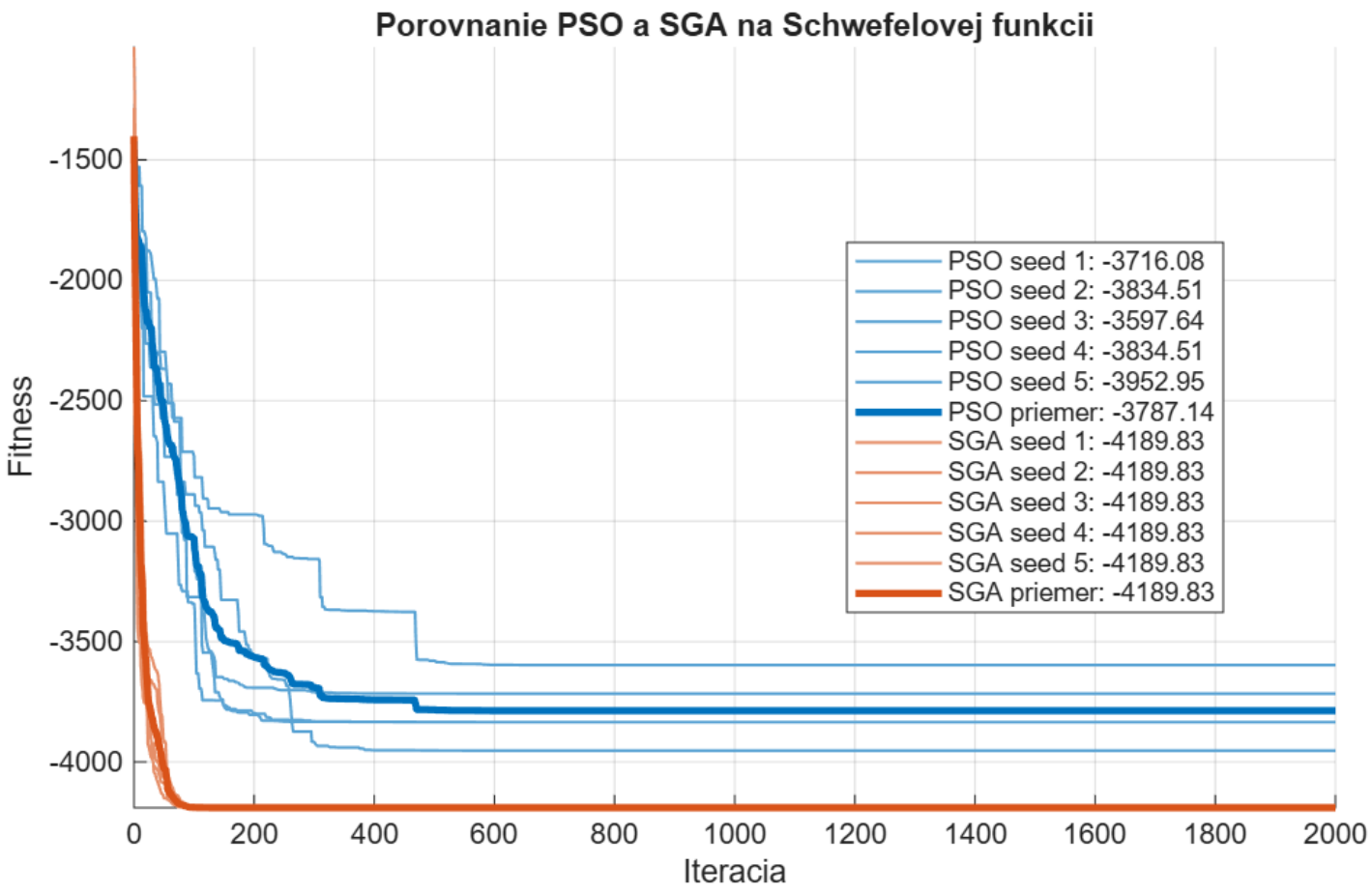
```
% Uvodne nastavenia
nvars = 10;                     % pocet premennych
Space = [-500*ones(1,nvars);
          500*ones(1,nvars)];   % definicny obor
runs = 5;                       % pocet behov
seeds = [1 2 3 4 5];            % rovnake seedy pre oba algoritmy
ngen = 2000;                    % pocet iteracii
popSize = 50;                   % pocet jedincov

% PSO
c0 = 0.85;                      % zotrvacnost castice
c1 = 1.5;                       % individualny faktor
c2 = 1.5;                       % socialny faktor
vMax = 200*ones(1,nvars);       % max. rychlost
```

```
% SGA
pmut_global = 0.3;
pmut_local = 0.3;
Amp_local = 5*ones(1,nvars);
sgaSize = 50;
bestNumsSGA = ones(1,2);
RandOldCountSGA = 10;
work1CountSGA = 18;
work2CountSGA = 20;
```

Výsledky:




Porovnanie PSO a SGA na Schwefelovej funkcii
Fitness
Iteracia
-1500
-2000
-2500
-3000
-3500
-4000
0
200
400
600
800
1000
1200
1400
1600
1800
2000
PSO seed 1: -3716.08
PSO seed 2: -3834.51
PSO seed 3: -3597.64
PSO seed 4: -3834.51
PSO seed 5: -3952.95
PSO priemer: -3787.14
SGA seed 1: -4189.83
SGA seed 2: -4189.83
SGA seed 3: -4189.83
SGA seed 4: -4189.83
SGA seed 5: -4189.83
SGA priemer: -4189.83